MODULE 3: VECTOR DIFFERENTIATION AND INTEGRATION (CO: 3)

Complete Study Textbook, Step-by-Step Derivations & VIT FAT Question Bank

Aligned with VIT BAMAT101 Official Syllabus, VIT Department of Mathematics Question Bank, and VIT FAT Past Papers

Scope of This Textbook (All 10 Syllabus Topics): This textbook covers every topic in Module 3 of your syllabus screenshot: (1) Scalar and vector valued functions—gradient, (2) Directional derivative, divergence and curl, (3) Scalar and vector potentials, (4) Line integral and problems, (5) Surface integral–volume integral and problems, (6) Statement of Green's theorem and problems, (7) Statement of Stokes' theorem and problems, (8) Statement of Gauss divergence theorem and problems, (9) Vector derivatives for electromagnetic simulation in computer aided design, and (10) Vector integrals in antenna design.

Contents

1 VIT FAT Exam Blueprint & Master Formula Sheet for Module 3 **2**

2 Topic 1: Scalar and Vector Valued Functions — Gradient **3**

2.1 Theory: Level Surfaces, Unit Normals, Tangent Planes & Angle Between Surfaces 3

3 Topic 2: Directional Derivative, Divergence, and Curl **3**

3.1 Theory & Vector Identities 3

3.2 Worked VIT Question Bank & Tutorial Problems (Topic 2) 4

4 Topic 3: Scalar and Vector Potentials **5**

4.1 Theory: Conservative Fields & Finding the Scalar Potential ϕ 5

4.2 Worked VIT Question Bank & FAT Problems (Topic 3) 5

5 Topic 4: Line Integrals and Problems **6**

5.1 Theory: Work Done & Path Independence 6

6 Topic 5: Surface Integrals, Volume Integrals, and Problems **6**

6.1 Theory: Projecting a Surface Integral Onto Coordinate Planes 6

6.2 Worked VIT Question Bank Problems (Topic 5) 6

7 Topic 6: Statement of Green's Theorem and Problems **8**

7.1 Statement of Green's Theorem in the Plane 8

7.2 Worked VIT Question Bank & Tutorial Problems (Topic 6) 8

8 Topic 7: Statement of Stokes' Theorem and Problems **9**

8.1 Statement of Stokes' Theorem 9

8.2 Worked VIT Question Bank & Tutorial Problems (Topic 7) 9

9 Topic 8: Statement of Gauss Divergence Theorem and Problems **10**

9.1 Statement of Gauss Divergence Theorem 10

9.2 Worked VIT Question Bank & Tutorial Problems (Topic 8) 10

10 Topic 9 & Topic 10: Electromagnetic Simulation in CAD & Antenna Design **11**

10.1 Topic 9: Vector Derivatives for Electromagnetic Simulation in CAD 11

10.2 Topic 10: Vector Integrals in Antenna Design 12

1 VIT FAT Exam Blueprint & Master Formula Sheet for Module 3

Module 3 is the highest-weighted module in VIT Calculus FAT papers (**20 to 30 marks**, typically one 10-mark question on Vector Differentiation/Scalar Potential and one or two 10-mark questions on Green's, Stokes', or Gauss Divergence Theorem).

Table 1: Vector Differential Operators & Fundamental Integral Theorems Master Sheet

Concept / Theorem	Core Formula	Key Physical / Exam Rule
Gradient $\nabla\phi$	$\hat{i}\frac{\partial\phi}{\partial x}+\hat{j}\frac{\partial\phi}{\partial y}+\hat{k}\frac{\partial\phi}{\partial z}$	Points in direction of max increase; normal to surface $\phi(x,y,z)=c$; unit normal $\hat{n}=\frac{\nabla\phi}{\lvert\nabla\phi\rvert}$.
Directional Derivative	$D_{\hat{u}}\phi=\nabla\phi\cdot\frac{\vec{a}}{\lvert\vec{a}\rvert}$	Rate of change of ϕ along $\vec{a}$. Maximum value is $\lvert\nabla\phi\rvert$.
Divergence $\nabla\cdot\vec{F}$	$\frac{\partial F_1}{\partial x}+\frac{\partial F_2}{\partial y}+\frac{\partial F_3}{\partial z}$	Scalar measure of outflow. If $\nabla\cdot\vec{F}=0$, $\vec{F}$ is **solenoidal** (incompressible).
Curl $\nabla\times\vec{F}$	$\begin{vmatrix}\hat{i}&\hat{j}&\hat{k}\\ \partial_x&\partial_y&\partial_z\\ F_1&F_2&F_3\end{vmatrix}$	Vector measure of rotation. If $\nabla\times\vec{F}=\vec{0}$, $\vec{F}$ is **irrotational / conservative** and $\vec{F}=\nabla\phi$.
Green's Theorem	$\oint_C(M\,dx+N\,dy)=\iint_R\left(\frac{\partial N}{\partial x}-\frac{\partial M}{\partial y}\right)dx\,dy$	Connects closed 2D line integral (counterclockwise) to double integral over planar region R.
Stokes' Theorem	$\oint_C\vec{F}\cdot d\vec{r}=\iint_S(\nabla\times\vec{F})\cdot\hat{n}\,dS$	Connects closed 3D curve C to open surface S bounded by C.
Gauss Divergence	$\iint_S\vec{F}\cdot\hat{n}\,dS=\iiint_V(\nabla\cdot\vec{F})\,dV$	Connects outward flux through closed surface S to triple integral over enclosed volume V.

[VIT FAT EXAM INSIGHT] "Verify" vs. "Evaluate" in VIT FAT Papers:

- If the question says **"Evaluate using Green's / Stokes' / Gauss Divergence Theorem"**, you only compute the **easier side** (usually the RHS double/triple integral) for full 10 marks!
- If the question says **"Verify Green's / Stokes' / Gauss Divergence Theorem"**, you **must** evaluate **both** the LHS and RHS independently and show that LHS = RHS (5 marks for LHS + 5 marks for RHS).

2 Topic 1: Scalar and Vector Valued Functions — Gradient

2.1 Theory: Level Surfaces, Unit Normals, Tangent Planes & Angle Between Surfaces

Let $\phi(x,y,z)$ be a differentiable scalar field. The vector differential operator **del** (∇) is $\nabla = \hat{i}\frac{\partial}{\partial x} + \hat{j}\frac{\partial}{\partial y} + \hat{k}\frac{\partial}{\partial z}$.

1. **Gradient**: $\operatorname{grad}\phi = \nabla\phi = \frac{\partial\phi}{\partial x}\hat{i} + \frac{\partial\phi}{\partial y}\hat{j} + \frac{\partial\phi}{\partial z}\hat{k}$.
2. **Unit Normal Vector** to the level surface $\phi(x,y,z) = c$ at point $P(x_0,y_0,z_0)$:
$$\hat{n} = \pm\frac{\nabla\phi|_P}{|\nabla\phi|_P|}.$$
3. **Equation of Tangent Plane** at $P(x_0,y_0,z_0)$: $(x-x_0)\phi_x(P) + (y-y_0)\phi_y(P) + (z-z_0)\phi_z(P) = 0$.
4. **Angle θ Between Two Surfaces** $\phi_1(x,y,z) = c_1$ and $\phi_2(x,y,z) = c_2$ at a common point P:
$$\cos\theta = \frac{\nabla\phi_1 \cdot \nabla\phi_2}{|\nabla\phi_1|\,|\nabla\phi_2|}. \qquad \text{(Surfaces are orthogonal if } \nabla\phi_1 \cdot \nabla\phi_2 = 0.\text{)}$$

Problem 3.1.1 (VIT FAT Standard Problem): *Find the angle between the surfaces $x^2+y^2+z^2 = 9$ and $z = x^2+y^2-3$ at the point $(2,-1,2)$.*

Solution:

Let $\phi_1 = x^2+y^2+z^2-9$ and $\phi_2 = x^2+y^2-z-3$. At $P(2,-1,2)$:

$$\nabla\phi_1 = 2x\hat{i} + 2y\hat{j} + 2z\hat{k}\Big|_{(2,-1,2)} = 4\hat{i} - 2\hat{j} + 4\hat{k}, \qquad |\nabla\phi_1| = \sqrt{16+4+16} = 6.$$

$$\nabla\phi_2 = 2x\hat{i} + 2y\hat{j} - \hat{k}\Big|_{(2,-1,2)} = 4\hat{i} - 2\hat{j} - \hat{k}, \qquad |\nabla\phi_2| = \sqrt{16+4+1} = \sqrt{21}.$$

Thus:

$$\cos\theta = \frac{(4)(4)+(-2)(-2)+(4)(-1)}{6\sqrt{21}} = \frac{16+4-4}{6\sqrt{21}} = \frac{16}{6\sqrt{21}} = \frac{8}{3\sqrt{21}} \implies \theta = \cos^{-1}\left(\frac{8}{3\sqrt{21}}\right). \quad \square$$

3 Topic 2: Directional Derivative, Divergence, and Curl

3.1 Theory & Vector Identities

1. **Directional Derivative**: The directional derivative of $\phi(x,y,z)$ at P in the direction of vector $\vec{a}$ is:
$$D_{\vec{a}}\phi = \nabla\phi|_P \cdot \frac{\vec{a}}{|\vec{a}|}.$$
The directional derivative is **maximum** in the direction of $\nabla\phi$, and its maximum value is $|\nabla\phi|$.
2. **Fundamental Vector Identities (Always True for Smooth Fields)**:
   - $\operatorname{curl}(\operatorname{grad}\phi) = \nabla\times(\nabla\phi) = \vec{0}$ (Every gradient field is irrotational).
   - $\operatorname{div}(\operatorname{curl}\vec{F}) = \nabla\cdot(\nabla\times\vec{F}) = 0$ (Every curl field is solenoidal).
   - $\nabla\times(\nabla\times\vec{F}) = \nabla(\nabla\cdot\vec{F}) - \nabla^2\vec{F}$.

3.2 Worked VIT Question Bank & Tutorial Problems (Topic 2)

Problem 3.2.1 (VIT Question Bank Q1 & Tutorial Q2): *(a) Find the directional derivative of $\phi(x,y,z) = x^2 - y^2 + 2z^2$ in the direction of $\vec{a} = 4\hat{i} - 2\hat{j} + \hat{k}$ at $(1,2,3)$. (b) Find the directional derivative of $\phi = xy + yz + zx$ at $(1,2,0)$ in the direction of $\hat{i} + 2\hat{j} + 2\hat{k}$ and its maximum value.*

Solution:

Part (a): $\nabla\phi = 2x\hat{i} - 2y\hat{j} + 4z\hat{k}$. At $(1,2,3)$, $\nabla\phi = 2\hat{i} - 4\hat{j} + 12\hat{k}$. The unit vector along $\vec{a}$ is $\hat{u} = \frac{4\hat{i}-2\hat{j}+\hat{k}}{\sqrt{16+4+1}} = \frac{4\hat{i}-2\hat{j}+\hat{k}}{\sqrt{21}}$.

$$D_{\hat{u}}\phi = (2\hat{i} - 4\hat{j} + 12\hat{k}) \cdot \frac{4\hat{i} - 2\hat{j} + \hat{k}}{\sqrt{21}} = \frac{8 + 8 + 12}{\sqrt{21}} = \frac{28}{\sqrt{21}} = \frac{4\sqrt{21}}{3}.$$

Part (b): $\nabla\phi = (y+z)\hat{i} + (x+z)\hat{j} + (x+y)\hat{k}$. At $(1,2,0)$, $\nabla\phi = 2\hat{i} + \hat{j} + 3\hat{k}$. Along $\vec{a} = \hat{i} + 2\hat{j} + 2\hat{k}$ ($|\vec{a}| = 3$):

$$D_{\hat{u}}\phi = (2\hat{i} + \hat{j} + 3\hat{k}) \cdot \frac{\hat{i} + 2\hat{j} + 2\hat{k}}{3} = \frac{2 + 2 + 6}{3} = \frac{10}{3}.$$

The **maximum value** of the directional derivative at $(1,2,0)$ is $|\nabla\phi| = \sqrt{2^2 + 1^2 + 3^2} = \sqrt{14}$. □

Problem 3.2.2 (VIT Question Bank Q2): *Find the directional derivative of $\phi = x^2yz + 4xz^2$ at $(1,-2,-1)$ in the direction of the normal to the surface $f(x,y,z) = x\log z - y^2 = -1$ at $(-1,2,1)$.*

Solution:

1. **Gradient of ϕ at $(1,-2,-1)$:**

$$\nabla\phi = (2xyz + 4z^2)\hat{i} + (x^2z)\hat{j} + (x^2y + 8xz)\hat{k}\Big|_{(1,-2,-1)} = (4+4)\hat{i} + (-1)\hat{j} + (-2-8)\hat{k} = 8\hat{i} - \hat{j} - 10\hat{k}.$$

2. **Normal direction to surface $f = x\log z - y^2$ at $(-1,2,1)$:**

$$\vec{n} = \nabla f\Big|_{(-1,2,1)} = (\log z)\hat{i} - 2y\hat{j} + \frac{x}{z}\hat{k}\Big|_{(-1,2,1)} = 0\hat{i} - 4\hat{j} - \hat{k}, \qquad |\vec{n}| = \sqrt{0 + 16 + 1} = \sqrt{17}.$$

3. **Directional derivative**:

$$D_{\hat{n}}\phi = (8\hat{i} - \hat{j} - 10\hat{k}) \cdot \frac{-4\hat{j} - \hat{k}}{\sqrt{17}} = \frac{0 + 4 + 10}{\sqrt{17}} = \frac{14}{\sqrt{17}}. \quad \square$$

Problem 3.2.3 (VIT Question Bank Q4, Q5 & Tutorial Q3): *(a) Find the constant a if $\vec{f} = (x+3y)\hat{i} + (y-2z)\hat{j} + (x-az)\hat{k}$ is solenoidal. (b) If $\vec{f} = (3x^2y - z)\hat{i} + (xz^3 + y^4)\hat{j} - 2x^3z^2\hat{k}$, find $\nabla \times (\nabla \times \vec{f})$.*

Solution:

Part (a): For a solenoidal field, $\nabla \cdot \vec{f} = 0$:

$$\frac{\partial}{\partial x}(x+3y) + \frac{\partial}{\partial y}(y-2z) + \frac{\partial}{\partial z}(x-az) = 1 + 1 - a = 0 \implies a = 2.$$

Part (b): First compute $\nabla \times \vec{f}$:

$$\nabla \times \vec{f} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \partial_x & \partial_y & \partial_z \\ 3x^2y - z & xz^3 + y^4 & -2x^3z^2 \end{vmatrix} = (-3xz^2)\hat{i} - (-6x^2z^2 + 1)\hat{j} + (z^3 - 3x^2)\hat{k}.$$

Now take the curl a second time:

$$\nabla \times (\nabla \times \vec{f}) = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \partial_x & \partial_y & \partial_z \\ -3xz^2 & 6x^2z^2 - 1 & z^3 - 3x^2 \end{vmatrix} = -12x^2z\hat{i} + (6x - 6xz)\hat{j} + 12xz^2\hat{k}. \quad \square$$

4 Topic 3: Scalar and Vector Potentials

4.1 Theory: Conservative Fields & Finding the Scalar Potential ϕ

A vector field $\vec{F} = F_1\hat{i} + F_2\hat{j} + F_3\hat{k}$ is **irrotational (conservative)** if and only if $\nabla \times \vec{F} = \vec{0}$, which means there exists a **scalar potential** $\phi(x, y, z)$ such that $\vec{F} = \nabla\phi$, i.e.:

$$\frac{\partial \phi}{\partial x} = F_1, \qquad \frac{\partial \phi}{\partial y} = F_2, \qquad \frac{\partial \phi}{\partial z} = F_3.$$

Fastest Exam Method to Find $\phi(x, y, z)$ (No Mistakes Guaranteed): Integrate each partial derivative with respect to its variable, and keep each **distinct term** only once:

$$\phi(x, y, z) = \int F_1\, dx \;\cup\; \int F_2\, dy \;\cup\; \int F_3\, dz + C \qquad \text{(combining shared terms without duplication).}$$

Similarly, if $\vec{B}$ is solenoidal ($\nabla \cdot \vec{B} = 0$), there exists a **vector potential** $\vec{A}$ such that $\vec{B} = \nabla \times \vec{A}$.

4.2 Worked VIT Question Bank & FAT Problems (Topic 3)

Problem 3.3.1 (VIT Question Bank Q8 & BMAT101L FAT Paper): *Find the constants a, b, c such that the vector field $\vec{F} = (x + 2y + az)\hat{i} + (bx - 3y - z)\hat{j} + (4x + cy + 2z)\hat{k}$ is irrotational. Hence find its scalar potential ϕ.*

Solution:

Step 1 (Irrotational condition $\nabla \times \vec{F} = \vec{0}$):

$$\nabla \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \partial_x & \partial_y & \partial_z \\ x + 2y + az & bx - 3y - z & 4x + cy + 2z \end{vmatrix} = (c - (-1))\hat{i} - (4 - a)\hat{j} + (b - 2)\hat{k} = \vec{0}.$$

Setting each component to zero:

$$c + 1 = 0 \implies c = -1, \qquad 4 - a = 0 \implies a = 4, \qquad b - 2 = 0 \implies b = 2.$$

Step 2 (Find the scalar potential ϕ): Substituting $a = 4, b = 2, c = -1$, we have $\vec{F} = (x + 2y + 4z)\hat{i} + (2x - 3y - z)\hat{j} + (4x - y + 2z)\hat{k}$. Integrating each component:

$$\int F_1\, dx = \int (x + 2y + 4z)\, dx = \frac{x^2}{2} + 2xy + 4xz,$$
$$\int F_2\, dy = \int (2x - 3y - z)\, dy = 2xy - \frac{3y^2}{2} - yz,$$
$$\int F_3\, dz = \int (4x - y + 2z)\, dz = 4xz - yz + z^2.$$

Combining each distinct term once:

$$\phi(x, y, z) = \frac{x^2}{2} - \frac{3y^2}{2} + z^2 + 2xy - yz + 4xz + k. \quad \square$$

Problem 3.3.2 (VIT Tutorial Sheet Q4): *Prove that $\vec{F} = (y^2 \cos x + z^3)\hat{i} + (2y \sin x - 4)\hat{j} + 3xz^2\hat{k}$ is irrotational and find its scalar potential.*

Solution:

1. **Check $\nabla \times \vec{F} = \vec{0}$:**

$$\nabla \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \partial_x & \partial_y & \partial_z \\ y^2 \cos x + z^3 & 2y \sin x - 4 & 3xz^2 \end{vmatrix} = (0 - 0)\hat{i} - (3z^2 - 3z^2)\hat{j} + (2y \cos x - 2y \cos x)\hat{k} = \vec{0}.$$

Hence $\vec{F}$ is irrotational. 2. **Scalar potential** ϕ:

$$\int (y^2 \cos x + z^3)\, dx = y^2 \sin x + xz^3, \quad \int (2y \sin x - 4)\, dy = y^2 \sin x - 4y, \quad \int 3xz^2\, dz = xz^3.$$

Combining unique terms gives $\phi(x, y, z) = y^2 \sin x + xz^3 - 4y + c$. □

5 Topic 4: Line Integrals and Problems

5.1 Theory: Work Done & Path Independence

For a curve C parameterized by $\vec{r}(t) = x(t)\hat{i} + y(t)\hat{j} + z(t)\hat{k}$ $(t_0 \le t \le t_1)$, the line integral of $\vec{F}$ along C (representing **work done** by force $\vec{F}$) is:

$$W = \int_C \vec{F} \cdot d\vec{r} = \int_C (F_1\, dx + F_2\, dy + F_3\, dz) = \int_{t_0}^{t_1} \left(F_1 \frac{dx}{dt} + F_2 \frac{dy}{dt} + F_3 \frac{dz}{dt} \right) dt.$$

Fundamental Theorem of Line Integrals: If $\vec{F} = \nabla\phi$ is conservative, then $\int_A^B \vec{F} \cdot d\vec{r} = \phi(B) - \phi(A)$ independent of the path!

Problem 3.4.1 (VIT Question Bank Q9): *Evaluate $\int_C \vec{f} \cdot d\vec{r}$ where $\vec{f} = 3xy\hat{i} - y^2\hat{j}$ and C is the part of the parabola $y = 2x^2$ from $(0, 0)$ to $(1, 2)$.*

Solution:

Along the parabola $y = 2x^2$, we have $dy = 4x\, dx$, and x varies from 0 to 1. Substituting $y = 2x^2$ and $dy = 4x\, dx$:

$$\int_C \vec{f} \cdot d\vec{r} = \int_0^1 \left[3x(2x^2) - (2x^2)^2(4x) \right] dx = \int_0^1 \left(6x^3 - 16x^5 \right) dx = \frac{3}{2} - \frac{8}{3} = -\frac{7}{6}. \quad \square$$

6 Topic 5: Surface Integrals, Volume Integrals, and Problems

6.1 Theory: Projecting a Surface Integral Onto Coordinate Planes

To evaluate the flux integral $\iint_S \vec{F} \cdot \hat{n}\, dS$ over a curved surface S given by $\phi(x, y, z) = c$:

1. Compute the unit normal $\hat{n} = \frac{\nabla\phi}{|\nabla\phi|}$ (pointing outward/upward in the first octant).
2. Project S onto one of the coordinate planes:
   - **Projection onto xy-plane** (R_{xy}): $dS = \dfrac{dx\, dy}{|\hat{n} \cdot \hat{k}|}$.
   - **Projection onto xz-plane** (R_{xz}): $dS = \dfrac{dx\, dz}{|\hat{n} \cdot \hat{j}|}$ (best for cylinders $x^2 + y^2 = a^2$!).
   - **Projection onto yz-plane** (R_{yz}): $dS = \dfrac{dy\, dz}{|\hat{n} \cdot \hat{i}|}$.

6.2 Worked VIT Question Bank Problems (Topic 5)

Problem 3.5.1 (VIT Question Bank Q11): *Evaluate $\iint_S \vec{F} \cdot \hat{n}\, dS$ where $\vec{F} = (x^2 + y^2)\hat{i} - 2x\hat{j} + 2yz\hat{k}$ and S is the surface of the plane $2x + y + 2z = 6$ in the first octant.*

Solution:

1. **Unit normal** $\hat{n}$: For $\phi = 2x + y + 2z = 6$, $\nabla\phi = 2\hat{i} + \hat{j} + 2\hat{k}$ and $|\nabla\phi| = \sqrt{4+1+4} = 3$. Thus $\hat{n} = \frac{2\hat{i}+\hat{j}+2\hat{k}}{3}$ and $\hat{n}\cdot\hat{k} = \frac{2}{3} \implies dS = \frac{dx\,dy}{2/3} = \frac{3}{2}dx\,dy$. 2. **Dot product** $\vec{F}\cdot\hat{n}$ **on the plane** $2z = 6-2x-y$:

$$\vec{F}\cdot\hat{n} = \frac{1}{3}\Big[2(x^2+y^2) - 2x + 2y(2z)\Big] = \frac{1}{3}\Big[2x^2+2y^2-2x+2y(6-2x-y)\Big] = \frac{2}{3}\left(x^2 - x + 6y - 2xy\right).$$

Notice how $2y^2$ cancels out cleanly! 3. **Integrate over the triangle** R_{xy} **in the** xy**-plane** ($2x + y \le 6$, so $0 \le x \le 3$ and $0 \le y \le 6-2x$):

$$\iint_S \vec{F}\cdot\hat{n}\,dS = \int_{x=0}^{3}\int_{y=0}^{6-2x} \frac{2}{3}(x^2-x+6y-2xy)\cdot\frac{3}{2}\,dy\,dx = \int_0^3 \Big[(x^2-x)y + (3-x)y^2\Big]_{y=0}^{y=6-2x}\,dx = 81. \quad \square$$

Problem 3.5.2 (VIT Question Bank Q12): *Evaluate* $\iint_S \vec{F}\cdot\hat{n}\,dS$ *where* $\vec{F} = z\hat{i} + x\hat{j} - 3y^2z\hat{k}$ *and* S *is the curved surface of the cylinder* $x^2+y^2 = 16$ *included in the first octant between* $z = 0$ *and* $z = 5$.

Solution:

1. **Unit normal** $\hat{n}$: For $\phi = x^2+y^2 = 16$, $\nabla\phi = 2x\hat{i} + 2y\hat{j}$ and $|\nabla\phi| = 2\sqrt{x^2+y^2} = 2(4) = 8$. Thus $\hat{n} = \frac{x\hat{i}+y\hat{j}}{4}$. 2. **Why we project onto the** xz**-plane (or** yz**-plane)**: Since the cylinder is vertical, $\hat{n}\cdot\hat{k} = 0$, so we cannot project onto the xy-plane! Projecting onto the xz-plane ($0 \le x \le 4$, $0 \le z \le 5$), we use $\hat{n}\cdot\hat{j} = \frac{y}{4}$, so $dS = \frac{dx\,dz}{y/4} = \frac{4}{y}dx\,dz$. 3. **Evaluate the integral**:

$$\vec{F}\cdot\hat{n} = \left(z\hat{i} + x\hat{j} - 3y^2z\hat{k}\right)\cdot\frac{x\hat{i}+y\hat{j}}{4} = \frac{xz+xy}{4}.$$

$$\iint_S \vec{F}\cdot\hat{n}\,dS = \int_{z=0}^{5}\int_{x=0}^{4}\left(\frac{xz+xy}{4}\right)\left(\frac{4}{y}\right)dx\,dz = \int_0^5\int_0^4\left(\frac{xz}{\sqrt{16-x^2}} + x\right)dx\,dz.$$

Since $\int_0^4 \frac{x}{\sqrt{16-x^2}}\,dx = \Big[-\sqrt{16-x^2}\Big]_0^4 = 4$ and $\int_0^4 x\,dx = 8$:

$$\iint_S \vec{F}\cdot\hat{n}\,dS = \int_0^5 (4z+8)\,dz = \Big[2z^2+8z\Big]_0^5 = 50+40 = 90. \quad \square$$

Problem 3.5.3 (VIT Question Bank Q10 & Q13 — Volume Integrals): *(a) If* $\vec{f} = (3x^2-2z)\hat{i} - 4xy\hat{j} - 5x\hat{k}$, *evaluate* $\iiint_V (\operatorname{curl}\vec{f})\,dV$ *where* V *is bounded by* $x=0, y=0, z=0$ *and* $3x+2y+3z=6$. *(b) If* $\vec{F} = (2x^2-3z)\hat{i} - 2xy\hat{j} - 4x\hat{k}$, *evaluate* $\iiint_V (\nabla\cdot\vec{F})\,dV$ *bounded by* $x=0, y=0, z=0$ *and* $2x+2y+z=4$.

Solution:

Part (a): Compute curl $\vec{f}$:

$$\nabla\times\vec{f} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \partial_x & \partial_y & \partial_z \\ 3x^2-2z & -4xy & -5x \end{vmatrix} = 0\hat{i} - (-5-(-2))\hat{j} + (-4y)\hat{k} = 3\hat{j} - 4y\hat{k}.$$

The volume of the tetrahedron $\frac{x}{2}+\frac{y}{3}+\frac{z}{2} \le 1$ is $V = \frac{1}{6}(2)(3)(2) = 2$, and integrating $-4y$ over the tetrahedron gives -6, yielding $6(\hat{j}-\hat{k})$ (or $6(-\hat{j}+\hat{k})$ when the plane is $3x+2y-3z=6$). **Part (b):** $\nabla\cdot\vec{F} = 4x - 2x + 0 = 2x$. Over $0 \le x \le 2$, $0 \le y \le 2-x$, $0 \le z \le 4-2x-2y$:

$$\iiint_V 2x\,dz\,dy\,dx = \int_0^2\int_0^{2-x} 2x(4-2x-2y)\,dy\,dx = \int_0^2 2x(2-x)^2\,dx = \frac{8}{3}. \quad \square$$

7 Topic 6: Statement of Green's Theorem and Problems

7.1 Statement of Green's Theorem in the Plane

If R is a closed, bounded region in the xy-plane bounded by a piecewise-smooth simple closed curve C oriented counterclockwise, and if $M(x,y)$ and $N(x,y)$ have continuous first partial derivatives on R, then:

$$\boxed{\oint_C (M\,dx + N\,dy) = \iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}\right) dx\,dy.}$$

7.2 Worked VIT Question Bank & Tutorial Problems (Topic 6)

Problem 3.6.1 (VIT Question Bank Q15 & Tutorial Q5): *Verify Green's theorem in the plane for $\oint_C (3x^2 - 8y^2)\,dx + (4y - 6xy)\,dy$, where C is the boundary of the triangle formed by $x = 0, y = 0$, and $x + y = 1$.*

Solution:

Here $M = 3x^2 - 8y^2$ and $N = 4y - 6xy$, so $\frac{\partial M}{\partial y} = -16y$ and $\frac{\partial N}{\partial x} = -6y$. **Part 1 (RHS — Double Integral over R):**

$$\text{RHS} = \iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}\right) dx\,dy = \int_0^1 \int_0^{1-x} 10y\,dy\,dx = \int_0^1 5(1-x)^2\,dx = \frac{5}{3}.$$

Part 2 (LHS — Line Integral along the 3 edges of triangle $O(0,0) \to A(1,0) \to B(0,1) \to O(0,0)$):

- **Along OA** $(y = 0, dy = 0, x : 0 \to 1)$: $\int_0^1 3x^2\,dx = \left[x^3\right]_0^1 = 1$.
- **Along AB** $(y = 1 - x, dy = -dx, x : 1 \to 0)$:

$$\int_1^0 \Big[(3x^2 - 8(1-x)^2) - (4(1-x) - 6x(1-x))\Big] dx = \int_0^1 \Big(11x^2 - 26x + 12\Big)\,dx = \left[\frac{11}{3} - 13 + 12\right] = \frac{8}{3}.$$

- **Along BO** $(x = 0, dx = 0, y : 1 \to 0)$: $\int_1^0 4y\,dy = \left[2y^2\right]_1^0 = -2$.

Summing the three segments: LHS $= 1 + \frac{8}{3} - 2 = \frac{5}{3} =$ RHS. Hence Green's theorem is verified! □

Problem 3.6.2 (VIT Question Bank Q16): *Verify Green's theorem for $\oint_C (xy + y^2)\,dx + x^2\,dy$, where C is the closed curve bounded by the parabola $y = x^2$ and the line $y = x$.*

Solution:

Here $M = xy + y^2 \implies \frac{\partial M}{\partial y} = x + 2y$, and $N = x^2 \implies \frac{\partial N}{\partial x} = 2x$. **Part 1 (RHS — Double Integral):**

$$\text{RHS} = \int_{x=0}^1 \int_{y=x^2}^x (2x - x - 2y)\,dy\,dx = \int_0^1 \Big[xy - y^2\Big]_{y=x^2}^{y=x} dx = \int_0^1 (x^4 - x^3)\,dx = \frac{1}{5} - \frac{1}{4} = -\frac{1}{20}.$$

Part 2 (LHS — Line Integral):

- **Curve C_1: Parabola $y = x^2$ $(dy = 2x\,dx)$ from $(0,0)$ to $(1,1)$:**

$$\int_{C_1} = \int_0^1 \Big[(x^3 + x^4) + x^2(2x)\Big]\,dx = \int_0^1 (3x^3 + x^4)\,dx = \frac{3}{4} + \frac{1}{5} = \frac{19}{20}.$$

- **Curve C_2: Line $y = x$ $(dy = dx)$ from $(1,1)$ to $(0,0)$:**

$$\int_{C_2} = \int_1^0 (2x^2 + x^2)\,dx = \Big[x^3\Big]_1^0 = -1.$$

Summing: LHS $= \frac{19}{20} - 1 = -\frac{1}{20} =$ RHS. Hence Green's theorem is verified! □

8 Topic 7: Statement of Stokes' Theorem and Problems

8.1 Statement of Stokes' Theorem

Let S be an oriented open surface bounded by a simple closed curve C with orientation consistent with the unit normal $\hat{n}$ via the right-hand rule. If $\vec{F}$ has continuous partial derivatives on S, then:

$$\boxed{\oint_C \vec{F} \cdot d\vec{r} = \iint_S (\nabla \times \vec{F}) \cdot \hat{n}\, dS.}$$

8.2 Worked VIT Question Bank & Tutorial Problems (Topic 7)

Problem 3.7.1 (VIT Question Bank Q18): *Verify Stokes' theorem for $\vec{F} = (x^2+y^2)\hat{i} - 2xy\hat{j}$ taken around the rectangle bounded by the lines $x = \pm a$, $y = 0$, $y = b$.*

Solution:

Since the rectangle lies in the xy-plane ($z = 0$), $\hat{n} = \hat{k}$ and $dS = dx\,dy$. 1. **RHS (Surface Integral)**:

$$\nabla \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \partial_x & \partial_y & \partial_z \\ x^2+y^2 & -2xy & 0 \end{vmatrix} = (-2y - 2y)\hat{k} = -4y\hat{k}.$$

$$\text{RHS} = \int_{x=-a}^{a} \int_{y=0}^{b} (-4y)\, dy\, dx = (2a)\left[-2y^2\right]_0^b = -4ab^2.$$

2. **LHS (Line Integral around $A(-a,0) \to B(a,0) \to C(a,b) \to D(-a,b) \to A(-a,0)$)**:

- AB $(y = 0, dy = 0, x : -a \to a)$: $\int_{-a}^{a} x^2\, dx = \frac{2a^3}{3}$.
- BC $(x = a, dx = 0, y : 0 \to b)$: $\int_0^b -2ay\, dy = -ab^2$.
- CD $(y = b, dy = 0, x : a \to -a)$: $\int_a^{-a} (x^2+b^2)\, dx = -\frac{2a^3}{3} - 2ab^2$.
- DA $(x = -a, dx = 0, y : b \to 0)$: $\int_b^0 2ay\, dy = -ab^2$.

Total LHS $= \frac{2a^3}{3} - ab^2 - \frac{2a^3}{3} - 2ab^2 - ab^2 = -4ab^2 =$ RHS. Verified! □

Problem 3.7.2 (VIT Question Bank Q19): *Verify Stokes' theorem for $\vec{F} = (2x-y)\hat{i} - yz^2\hat{j} - y^2z\hat{k}$ over the upper-half surface of the sphere $x^2+y^2+z^2 = 1$, bounded by its projection on the xy-plane.*

Solution:

1. **LHS (Line Integral along boundary circle $C : x^2+y^2 = 1, z = 0, dz = 0$)**: On $z = 0$, $\vec{F} = (2x-y)\hat{i}$. Parameterizing $x = \cos t, y = \sin t$ $(0 \le t \le 2\pi)$:

$$\oint_C \vec{F} \cdot d\vec{r} = \oint_C (2x - y)\, dx = \int_0^{2\pi} (2\cos t - \sin t)(-\sin t)\, dt = \int_0^{2\pi} \left(-\sin 2t + \sin^2 t\right) dt = 0 + \pi = \pi.$$

2. **RHS (Surface Integral over upper hemisphere S)**:

$$\nabla \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \partial_x & \partial_y & \partial_z \\ 2x-y & -yz^2 & -y^2z \end{vmatrix} = (-2yz - (-2yz))\hat{i} - 0\hat{j} + (0 - (-1))\hat{k} = \hat{k}.$$

Projecting S onto the unit disk R_{xy} $(x^2+y^2 \le 1)$ in the xy-plane, where $dS = \frac{dx\,dy}{\hat{n}\cdot\hat{k}}$:

$$\iint_S (\nabla \times \vec{F}) \cdot \hat{n}\, dS = \iint_S (\hat{k}\cdot\hat{n}) \frac{dx\,dy}{\hat{n}\cdot\hat{k}} = \iint_{x^2+y^2\le 1} dx\,dy = \text{Area of unit circle} = \pi.$$

Thus LHS = RHS = π. Verified! □

Problem 3.7.3 (VIT Question Bank Q20): *Evaluate $\oint_C \vec{F} \cdot d\vec{r}$ by Stokes' theorem where $\vec{F} = y^2\hat{i} + x^2\hat{j} - (x+z)\hat{k}$ and C is the boundary of the triangle with vertices $(0,0,0), (1,0,0), (1,1,0)$.*

Solution:

The triangle lies in the xy-plane ($z = 0, \hat{n} = \hat{k}, dS = dx\,dy$) bounded by $0 \le x \le 1$ and $0 \le y \le x$. Computing the curl: $\nabla \times \vec{F} = 0\hat{i} + \hat{j} + (2x - 2y)\hat{k}$, so $(\nabla \times \vec{F}) \cdot \hat{k} = 2(x - y)$. By Stokes' theorem:

$$\oint_C \vec{F} \cdot d\vec{r} = \int_{x=0}^{1} \int_{y=0}^{x} 2(x-y)\,dy\,dx = \int_0^1 \Big[2xy - y^2\Big]_{y=0}^{y=x}\,dx = \int_0^1 x^2\,dx = \frac{1}{3}. \quad \square$$

9 Topic 8: Statement of Gauss Divergence Theorem and Problems

9.1 Statement of Gauss Divergence Theorem

Let V be a closed three-dimensional region bounded by a piecewise-smooth closed surface S with outward-pointing unit normal $\hat{n}$. If $\vec{F}$ has continuous first partial derivatives on V, then:

$$\boxed{\iint_S \vec{F} \cdot \hat{n}\,dS = \iiint_V (\nabla \cdot \vec{F})\,dV.}$$

9.2 Worked VIT Question Bank & Tutorial Problems (Topic 8)

Problem 3.8.1 (VIT Question Bank Q17): *Using Gauss divergence theorem, evaluate $\iint_S \vec{F} \cdot \hat{n}\,dS$ where $\vec{F} = 2xy\hat{i} + yz^2\hat{j} + xz\hat{k}$ and S is the surface of the parallelepiped $0 \le x \le 2,\ 0 \le y \le 1,\ 0 \le z \le 3$.*

Solution:

Compute the divergence $\nabla \cdot \vec{F}$:

$$\nabla \cdot \vec{F} = \frac{\partial}{\partial x}(2xy) + \frac{\partial}{\partial y}(yz^2) + \frac{\partial}{\partial z}(xz) = 2y + z^2 + x.$$

By Gauss Divergence Theorem:

$$\iint_S \vec{F} \cdot \hat{n}\,dS = \int_0^2 \int_0^1 \int_0^3 (x + 2y + z^2)\,dz\,dy\,dx = \int_0^2 \int_0^1 (3x + 6y + 9)\,dy\,dx = \int_0^2 (3x + 12)\,dx = 30. \quad \square$$

Problem 3.8.2 (VIT Tutorial Sheet Q10 — Full Verification on a Box): *Verify the divergence theorem for $\vec{F} = (x^2 - yz)\hat{i} + (y^2 - zx)\hat{j} + (z^2 - xy)\hat{k}$ taken over the rectangular parallelepiped $0 \le x \le a,\ 0 \le y \le b,\ 0 \le z \le c$.*

Solution:

1. **RHS (Volume Integral)**:

$$\nabla \cdot \vec{F} = 2x + 2y + 2z = 2(x + y + z).$$

$$\text{RHS} = \int_0^a \int_0^b \int_0^c 2(x + y + z)\,dz\,dy\,dx = 2\left(\frac{a^2}{2}bc + a\frac{b^2}{2}c + ab\frac{c^2}{2}\right) = abc(a + b + c).$$

2. **LHS (Surface Integral over the 6 faces)**:

- **Pair $x = a$ ($\hat{n} = \hat{i}$) and $x = 0$ ($\hat{n} = -\hat{i}$)**:

$$\int_0^b \int_0^c \Big[(a^2 - yz) - (0 - yz)\Big]\,dz\,dy = \int_0^b \int_0^c a^2\,dz\,dy = a^2bc.$$

- **Pair $y = b$ ($\hat{n} = \hat{j}$) and $y = 0$ ($\hat{n} = -\hat{j}$):** By symmetry, $\int_0^a \int_0^c b^2\, dz\, dx = ab^2c$.
- **Pair $z = c$ ($\hat{n} = \hat{k}$) and $z = 0$ ($\hat{n} = -\hat{k}$):** By symmetry, $\int_0^a \int_0^b c^2\, dy\, dx = abc^2$.

Summing all 6 faces: LHS $= a^2bc + ab^2c + abc^2 = abc(a+b+c) =$ RHS. Verified! □

Problem 3.8.3 (VIT Tutorial Sheet Q9): *Evaluate $\iint_S (ax\hat{i}+by\hat{j}+cz\hat{k})\cdot\hat{n}\, dS$ (or $\iint_S (ax^2+by^2+cz^2)\, dS$ on unit sphere $x^2+y^2+z^2=1$) using the Divergence theorem.*

Solution:

On the unit sphere $x^2+y^2+z^2=1$, the outward unit normal is $\hat{n} = x\hat{i}+y\hat{j}+z\hat{k}$. Writing $ax^2+by^2+cz^2 = (ax\hat{i}+by\hat{j}+cz\hat{k})\cdot(x\hat{i}+y\hat{j}+z\hat{k}) = \vec{F}\cdot\hat{n}$ where $\vec{F} = ax\hat{i}+by\hat{j}+cz\hat{k}$:

$$\nabla\cdot\vec{F} = a+b+c.$$

By Gauss Divergence Theorem:

$$\iint_S \vec{F}\cdot\hat{n}\, dS = \iiint_V (a+b+c)\, dV = (a+b+c)\cdot(\text{Volume of unit sphere}) = \frac{4\pi}{3}(a+b+c). \quad \square$$

10 Topic 9 & Topic 10: Electromagnetic Simulation in CAD & Antenna Design

10.1 Topic 9: Vector Derivatives for Electromagnetic Simulation in CAD

Modern electromagnetic Computer-Aided Design (CAD) software (such as ANSYS HFSS and CST Studio) solves **Maxwell's Equations** in differential form across 3D tetrahedral meshes:

Table 2: Maxwell's Equations in Differential & Integral Forms for CAD Simulation

Law	Differential Form	Integral Form	Vector Theorem Bridge
Gauss's Law (E)	$\nabla\cdot\vec{D} = \rho_v$	$\iint_S \vec{D}\cdot\hat{n}\, dS = \iiint_V \rho_v\, dV$	Gauss Divergence Theorem
Gauss's Law (B)	$\nabla\cdot\vec{B} = 0$	$\iint_S \vec{B}\cdot\hat{n}\, dS = 0$	Magnetic field $\vec{B}$ is solenoidal ($\vec{B} = \nabla\times\vec{A}$)
Faraday's Law	$\nabla\times\vec{E} = -\frac{\partial\vec{B}}{\partial t}$	$\oint_C \vec{E}\cdot d\vec{r} = -\frac{d}{dt}\iint_S \vec{B}\cdot\hat{n}\, dS$	Stokes' Theorem (Electrostatics: $\nabla\times\vec{E} = \vec{0} \implies \vec{E} = -\nabla V$)
Ampère–Maxwell	$\nabla\times\vec{H} = \vec{J} + \frac{\partial\vec{D}}{\partial t}$	$\oint_C \vec{H}\cdot d\vec{r} = \iint_S \left(\vec{J} + \frac{\partial\vec{D}}{\partial t}\right)\cdot\hat{n}\, dS$	Stokes' Theorem

Problem 3.9.1 (Applied EM-CAD Problem): *In a 3D CAD electrostatic simulation, the electric potential in a region is $V(x,y,z) = 3x^2y - yz^2$ Volts. Find (i) the electric field vector $\vec{E} = -\nabla V$ at $(1,2,-1)$, and (ii) the volume charge density $\rho_v = \varepsilon_0(\nabla\cdot\vec{E})$ at that point.*

Solution:

1. **Electric field $\vec{E} = -\nabla V$:**

$$\vec{E} = -\left(6xy\hat{i} + (3x^2 - z^2)\hat{j} - 2yz\hat{k}\right). \quad \text{At } (1,2,-1): \vec{E} = -12\hat{i} - 2\hat{j} - 4\hat{k} \text{ V/m}.$$

2. **Charge density** $\rho_v = \varepsilon_0(\nabla \cdot \vec{E}) = -\varepsilon_0 \nabla^2 V$:

$$\nabla \cdot \vec{E} = \frac{\partial}{\partial x}(-6xy) + \frac{\partial}{\partial y}(z^2 - 3x^2) + \frac{\partial}{\partial z}(2yz) = -6y + 0 + 2y = -4y.$$

At $(1, 2, -1)$, $\rho_v = -4(2)\varepsilon_0 = -8\varepsilon_0$ C/m^3. □

10.2 Topic 10: Vector Integrals in Antenna Design

In antenna engineering, vector integrals govern both radiation generation and far-field power measurement:

1. **Magnetic Vector Potential of a Wire Antenna (Line Integral)**: For a thin wire antenna along curve C carrying current $I(z')$, the vector potential at distance R is $\vec{A}(\vec{r}) = \frac{\mu_0}{4\pi} \int_C \frac{I(\vec{r}\,')e^{-ikR}}{R}\, d\vec{r}\,'$.
2. **Total Radiated Power via Poynting Vector Flux (Surface Integral)**: Given time-averaged Poynting vector $\vec{S}_{\text{avg}} = \frac{1}{2}\,\text{Re}(\vec{E} \times \vec{H}^*)$, the total power P_{rad} radiated by an antenna through a far-field sphere S of radius r is:

$$P_{\text{rad}} = \iint_S \vec{S}_{\text{avg}} \cdot \hat{n}\, dS = \int_{\theta=0}^{2\pi} \int_{\phi=0}^{\pi} S_r(r, \phi, \theta)\, r^2 \sin\phi\, d\phi\, d\theta.$$

Problem 3.10.1 (Applied Antenna Design Problem): *A half-wave dipole antenna centered at the origin radiates a far-field radial Poynting power density approximated by* $\vec{S} = \frac{P_0 \sin^2\phi}{r^2}\hat{e}_r$ *(where* $\phi \in [0, \pi]$ *is the polar angle from the z-axis). Evaluate the surface integral* $P_{rad} = \iint_S \vec{S} \cdot \hat{n}\, dS$ *over a sphere of radius* r *to find the total radiated power.*

Solution:

On the sphere of radius r, $\hat{n} = \hat{e}_r$, $\vec{S} \cdot \hat{n} = \frac{P_0 \sin^2\phi}{r^2}$, and the spherical surface element is $dS = r^2 \sin\phi\, d\phi\, d\theta$:

$$P_{\text{rad}} = \int_0^{2\pi} \int_0^{\pi} \left(\frac{P_0 \sin^2\phi}{r^2}\right) r^2 \sin\phi\, d\phi\, d\theta = 2\pi P_0 \int_0^{\pi} \sin^3\phi\, d\phi = 2\pi P_0 \left(\frac{4}{3}\right) = \frac{8\pi P_0}{3}. \quad \square$$